

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

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Abstract—Climate change and rising industrial carbon emissions have increased the need for transparent and intelligent carbon credit trading systems. Existing carbon markets suffer from issues such as delayed verification, lack of transparency, static pricing mechanisms, and weak fraud detection methods. These limitations reduce trust, efficiency, and scalability in carbon trading ecosystems. This paper proposes an AI-governed carbon credit exchange integrated with a Digital Carbon Passport to enable secure, transparent, and intelligent carbon trading. The proposed system combines artificial intelligence, dynamic pricing, secure ledger verification, trust scoring, and cryptographic transaction protection to improve decision-making and system reliability. The platform introduces a Secure Trade Authorization and Verification Pipeline (STAVP) incorporating C³T obfuscation, AES-256 encryption, and SHA-256 hashing to ensure tamper-proof transactions. The system was evaluated using simulated market environments involving concurrent transactions, dynamic pricing scenarios, and multi-role interactions. Experimental results demonstrate improved transparency, enhanced fraud detection capability, secure transaction processing, and adaptive pricing performance compared to conventional systems. The proposed framework provides a scalable and intelligent foundation for future sustainable carbon trading ecosystems.

Keywords—Artificial Intelligence, Carbon Credit Trading, Digital Carbon Passport, Dynamic Pricing, Blockchain Security, Carbon Market, Fraud Detection, Sustainable Computing.

I. INTRODUCTION

Climate change has become one of the most significant global challenges affecting environmental sustainability and economic stability. Industrial carbon emissions continue to

increase, creating an urgent need for efficient carbon management systems. Carbon credit trading has emerged as a market-driven mechanism that encourages industries to reduce emissions by enabling organizations with surplus carbon credits to trade with organizations exceeding emission limits.

Although carbon markets provide economic incentives for sustainability, existing systems suffer from multiple operational and technical limitations. Most current platforms rely heavily on manual verification processes, centralized decision-making, delayed audits, and static pricing models. These approaches often result in low transparency, inefficient processing, weak scalability, and higher risks of fraud or manipulation.

Recent advancements in artificial intelligence and secure digital systems offer opportunities to modernize carbon trading infrastructures. Artificial intelligence enables intelligent validation, anomaly detection, predictive analysis, and adaptive pricing. At the same time, secure cryptographic frameworks can improve transaction integrity and reduce the possibility of tampering or double-counting.

To address these challenges, this paper proposes an AI-governed carbon credit exchange integrated with a Digital Carbon Passport framework. The Digital Carbon Passport acts as a continuously updated digital identity containing organizational emission history, compliance records, sustainability performance, and transaction activities. This enables transparent monitoring and intelligent governance throughout the carbon trading lifecycle.

The proposed system integrates several advanced components, including AI-based dynamic pricing mechanisms, real-time validation and anomaly detection, trust score evaluation, secure cryptographic transaction processing, and role-based monitoring dashboards. A layered architecture and Secure Trade Authorization and Verification Pipeline

(STAVP) are implemented to ensure secure and scalable transaction handling. The framework incorporates C³T obfuscation, AES-256 encryption, and SHA-256 hashing to provide confidentiality, integrity, and tamper resistance.

The major contributions of this paper are:

- Introduction of a Digital Carbon Passport for transparent carbon tracking.
- Development of AI-based dynamic carbon credit pricing.
- Integration of trust-based governance and anomaly detection.
- Design of a secure multi-layer cryptographic transaction framework.
- Creation of a scalable architecture supporting real-time monitoring and intelligent decision-making.

The remainder of the paper is organized as follows: Section II discusses related literature and existing research. Section III explains the proposed methodology and system design. Section IV presents implementation details and experimental evaluation. Section V discusses results and analysis, while Section VI concludes the paper and outlines future work.

II. LITERATURE REVIEW

Recent advancements in carbon credit trading systems have increasingly focused on integrating blockchain, artificial intelligence, and quantum computing technologies to improve transparency, efficiency, and sustainability. Golshani and Mankar [1] proposed a democratized blockchain-based carbon credit trading platform that enhances transparency and enables secure peer-to-peer transactions. Their work highlights the importance of decentralized systems in reducing intermediary dependency and improving trust among stakeholders.

Saeed, Wen, and Afzal [2] introduced a unified FLC-blockchain framework for optimized carbon credit trading in multi-microgrid environments. The study demonstrated that combining fuzzy logic control with blockchain improves decision-making accuracy and energy trading efficiency. This framework also supports secure and automated trading operations within distributed energy systems.

Research on quantum technologies in the energy sector has gained significant attention in recent years. Strata et al. [3] conducted a systematic review on quantum machine learning applications for the energy industry and identified its potential in optimization, forecasting, and energy management problems. Similarly, Tripathi, Upadhyay, and Soni [4] analyzed the trade-offs between Quantum Neural Networks

(QNN) and Quantum Long Short-Term Memory (QLSTM) models, emphasizing their energy-efficient computation capabilities for sustainable AI applications.

Sinha et al. [5] compared quantum and classical algorithms for renewable energy optimization and concluded that quantum-based methods can provide faster and more optimized solutions for complex energy management problems. Their findings suggest that quantum computing can significantly contribute to future sustainable energy systems and carbon optimization strategies.

Blockchain-enabled peer-to-peer energy and carbon trading frameworks have also been widely explored. Boumaiza [6] proposed a blockchain-centric P2P trading framework that integrates both carbon and energy trades. The framework ensures transparency, secure transactions, and improved energy resource utilization among consumers and producers.

Machine learning techniques have been incorporated into blockchain-based carbon systems to address fraud and greenwashing issues. Raja Segaran [7] developed an efficient machine learning model integrated with blockchain technology to detect suspicious carbon trading activities and improve system reliability. The study emphasized the importance of intelligent verification mechanisms in carbon markets.

Tokenization and digital asset management have emerged as important trends in sustainable finance. Tlili [8] discussed the role of blockchain in tokenized carbon markets and explained how digital carbon assets can improve liquidity, accessibility, and market participation. The study also highlighted the growing adoption of decentralized finance mechanisms in environmental sustainability initiatives.

Zhang, Chen, and Yue [9] analyzed the implementation of blockchain technology in the Shanghai Environment and Energy Exchange. Their case study demonstrated that blockchain improves traceability, transparency, and operational efficiency in carbon trading markets while reducing the possibility of fraudulent transactions.

Furthermore, Olajide et al. [10] explored the combined role of artificial intelligence and blockchain in enhancing modern carbon markets. Their research showed that fintech innovations can automate carbon credit verification, optimize trading strategies, and improve market efficiency through intelligent analytics and decentralized infrastructures.

Overall, the reviewed literature indicates that blockchain technology provides transparency and security in carbon trading, while artificial intelligence and quantum computing contribute advanced optimization and predictive capabilities.

However, existing systems still face challenges related to scalability, interoperability, fraud prevention, and real-time optimization. These research gaps motivate the development of more intelligent, secure, and sustainable carbon credit trading frameworks.

III. PROPOSED METHODOLOGY

The proposed system introduces an AI-governed carbon credit exchange integrated with a Digital Carbon Passport to improve transparency, security, scalability, and intelligent decision-making in carbon trading ecosystems. The methodology combines artificial intelligence, secure cryptographic verification, trust evaluation, and dynamic pricing mechanisms into a unified framework capable of supporting real-time monitoring and automated governance.

The system follows a layered architecture in which each module performs dedicated operations while securely interacting with other system components. The complete workflow begins with emission data collection and validation, followed by trust score computation, AI-based dynamic pricing, secure trade execution, and immutable ledger storage. The proposed methodology aims to overcome the limitations of conventional carbon markets such as delayed verification, lack of transparency, static pricing, and weak fraud detection mechanisms.

A. System Workflow

The system begins by collecting emission and sustainability data from organizations. AI validation models analyze the submitted information to identify inconsistencies, abnormal patterns, and compliance violations. Validated information is stored within the Digital Carbon Passport, which continuously maintains organizational environmental records.



Fig. 1. The sequential verification and trade execution workflow.

The AI pricing engine then calculates adaptive carbon credit prices based on market demand, credit availability, organizational trust score, emission performance, and market volatility. Transactions are securely processed through the Secure Trade Authorization and Verification Pipeline

(STAVP), where cryptographic security layers protect data before permanent ledger storage.

B. Digital Carbon Passport

The Digital Carbon Passport acts as a continuously updated environmental identity for organizations. It stores emission history, compliance records, sustainability metrics, transaction history, trust scores, and carbon credit balance. The passport improves transparency, accountability, and auditability within the carbon trading ecosystem.

C. AI-Based Dynamic Pricing

The AI pricing engine dynamically calculates carbon credit values using market and environmental conditions. The pricing mechanism considers the supply-demand ratio, trust score, emission reduction performance, and market volatility. The pricing equation is defined as:

$$Price = Base \times Demand \times TrustAdjustment \times VolatilityModifier \quad (1)$$

This allows real-time adaptive pricing instead of fixed traditional pricing models.

D. Trust Score Evaluation

The trust score mechanism evaluates organizational credibility using multiple weighted parameters, including compliance consistency, sustainability performance, reporting accuracy, and trading history. The trust score equation is:

$$TrustScore = (0.30 \times Compliance) + (0.25 \times Sustainability) + (0.25 \times Reporting) + (0.20 \times Trading) \quad (2)$$

Organizations with higher trust scores receive improved market reliability and trading advantages.

E. Secure Trade Authorization and Verification Pipeline (STAVP)

The STAVP module ensures secure transaction processing through multiple stages: Data ingestion, AI validation, Trust evaluation, Dynamic pricing, Trade execution, C³T obfuscation, AES-256 encryption, SHA-256 hashing, and Immutable ledger storage. This layered workflow prevents unauthorized access and data tampering.

F. Cryptographic Security Framework

The proposed framework integrates C³T Cipher Obfuscation, AES-256-GCM Encryption, and SHA-256 Hashing. AES encryption ensures confidentiality, SHA-256 guarantees integrity, and C³T obfuscation improves resistance against pattern-based attacks.

G. Role-Based Access Control (RBAC)

The RBAC system manages permissions for Company Administrators, Regulators, Auditors, and Market Operators. JWT-based authentication ensures secure user identity verification.

H. Layered System Architecture

The architecture consists of User Interface Layer, API Gateway Layer, Market Orchestrator Layer, AI and Trust Layer, Secure Ledger Layer, and Database Layer. This modular architecture improves scalability, security, and maintainability.

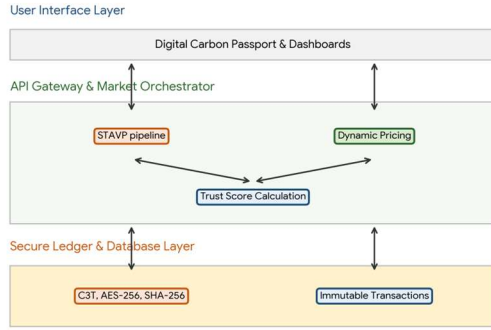


Fig. 2. High-level layered architecture of the AI-governed carbon credit exchange.

IV. RESULTS AND DISCUSSION

The proposed platform was successfully implemented and evaluated for secure and intelligent carbon credit trading. The system integrates AI-driven pricing, fraud detection, secure ledger verification, and real-time regulatory monitoring within a unified digital environment. The AI-based pricing model generated reliable carbon credit price predictions using market demand, supply, volatility, and trust score as input parameters. The implemented model achieved an R^2 value of 0.87 with a Mean Absolute Error (MAE) of \$1.23 per credit, indicating stable prediction performance.

The fraud detection subsystem demonstrated effective performance during evaluation. The system achieved an accuracy of 97.4%, precision of 96.1%, recall of 94.8%, and F1-score of 95.4%, confirming the reliability of the implemented anomaly detection and transaction verification mechanisms. The platform also maintained efficient computational performance with an average inference speed of 12.3 ms and a throughput of approximately 4,820 requests per second.

TABLE I

PERFORMANCE METRICS EVALUATION

Metric	Value
Accuracy	97.4%
Precision	96.1%
Recall	94.8%
F1-Score	95.4%
Inference Speed	12.3 ms
Throughput	4,820 req/s

The implemented STAVP hash-chain verification mechanism successfully ensured transaction integrity and prevented unauthorized modifications during trading operations. The regulator dashboard provided real-time monitoring through analytical visualization, KPI tracking, and fraud alerts, thereby improving transparency and regulatory oversight within the carbon credit ecosystem.

V. CONCLUSION AND FUTURE WORK

This paper presented an AI-governed carbon credit exchange integrated with a Digital Carbon Passport to address major challenges in existing carbon trading systems such as limited transparency, delayed verification, static pricing mechanisms, and centralized decision-making. The proposed framework combines Artificial Intelligence, trust-based evaluation, dynamic pricing, and secure cryptographic transaction management within a unified architecture to create a more intelligent and reliable carbon trading ecosystem.

The developed system successfully incorporates real-time emission validation, AI-driven anomaly detection, organizational trust scoring, and adaptive carbon credit pricing to improve operational efficiency and market responsiveness. In addition, the Secure Trade Authorization and Verification Pipeline (STAVP), integrated with C³T obfuscation, AES-256 encryption, and SHA-256 hashing, ensures confidentiality, integrity, and tamper resistance throughout the transaction lifecycle. The Digital Carbon Passport further enhances transparency and traceability by maintaining continuous records of emission history, compliance metrics, and trading activities.

Simulation and verification results demonstrate that the proposed framework is capable of handling concurrent transactions, maintaining secure data flow, and supporting scalable market operations while preserving transactional integrity and governance transparency. The layered architecture and modular design also improve maintainability and flexibility for future expansion.

Although the proposed system was evaluated in a controlled simulation environment using standards-driven synthetic datasets, the framework establishes a strong foundation for

future intelligent carbon trading platforms. Future work may focus on large-scale real-world deployment, integration with blockchain-based distributed ledgers, incorporation of advanced machine learning models for predictive analytics, and interoperability with national and international carbon market infrastructures. Additional research may also explore integration with IoT-enabled environmental monitoring systems and decentralized governance mechanisms to further enhance scalability, automation, and regulatory compliance.

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